

SRMS-004 — Small but Decisive Structural Deltas

Structural Recognition above Metric Similarity (SRMS)

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Repository: <https://github.com/sizhet/Structural-Recognition-above-Metric-Similarity>

Abstract

Modern AI systems are remarkably effective at recognizing broad semantic similarity.

However, many real-world decisions depend on structural differences that are extremely small in size but decisive in consequence.

These differences often occupy only a tiny fraction of the overall representation while determining legality, safety, correctness, executability, compatibility, or runtime behavior.

This paper introduces the concept of **Small but Decisive Structural Deltas**.

Rather than measuring how much an object differs, structural recognition should identify **which differences determine the decision boundary**.

Recognizing these decisive structural deltas is one of the central capabilities required for reliable AI.

1. Not All Structural Differences Matter Equally

Every complex object contains countless differences.

Most are insignificant.

Only a few determine meaning.

For example:

Program A

```
if (temperature > 80)
```

Program B

```
if (temperature >= 80)
```

The structural difference is only one character.

The runtime behavior may change completely.

The important question is therefore not:

How many differences exist?

The important question is:

Which difference determines the outcome?

2. Decisive Structural Deltas

A decisive structural delta is a structural difference that changes one or more of the following:

- semantic identity,
- execution path,
- decision boundary,
- safety condition,
- legal interpretation,
- runtime invariant,
- certification status,
- functional role.

These deltas are usually:

- sparse,
- discrete,
- localized,
- high-impact.

Their physical size is often tiny.

Their semantic influence is often enormous.

3. Typical Categories of Decisive Deltas

Although decisive deltas appear in many domains, they often belong to recurring structural families.

Logical Operators

Examples:

and

or

not

(remove "not")

if

unless

One logical operator may reverse the entire meaning.

Comparison Operators

Examples:

>

>=

<

<=

==

!=

One comparison operator may move execution across a boundary.

Boundary Conditions

Examples:

temperature = 100

versus

temperature >= 100

The structural change is minimal.

The activated runtime may be completely different.

State Changes

Examples:

enabled

disabled

normal

emergency

One state transition may activate an entirely different Calling Graph.

Scope

Examples:

global

local

public

private

inside

outside

Scope determines where a statement is valid.

Exceptions

Examples:

except

unless

only if

Exceptions often override an otherwise dominant structural base.

Certification

Examples:

verified

unverified

approved

unapproved

Two objects may be nearly identical while belonging to different certification classes.

4. Why Similarity Often Misses Them

Embedding models primarily represent the dominant structural base.

The decisive delta contributes only a small amount to the overall vector.

Consequently,

two objects may remain extremely close in embedding space even when they belong to opposite decision classes.

This is not necessarily an embedding error.

It is a consequence of averaging a very small but decisive structure into a much larger common context.

Similarity therefore observes:

Almost identical.

Reality observes:

Completely different.

5. Decision Boundaries Are Structural

Many engineering systems are organized around structural boundaries.

Examples include:

Safe

Unsafe

Executable

Non-Executable

Certified

Not Certified

Authorized

Unauthorized

Compatible

Incompatible

Crossing such a boundary may require changing only one structural element.

Recognition must therefore identify boundary-producing deltas rather than merely measuring global similarity.

6. Structural Recognition Must Detect Five Properties

A useful recognition system should determine:

Presence

Does the decisive delta exist?

Location

Where is it located?

Function

What role does it play?

Boundary

Which decision boundary does it control?

Consequence

What runtime behavior changes if the delta changes?

Only after answering these questions does similarity become reliable.

7. Relation to Recognition-Gated Cosine Scoring

Recognition-Gated Cosine Scoring relies on identifying decisive structural deltas before metric ranking.

Recognition therefore answers:

Which structural differences matter?

Cosine similarity answers:

Among structurally compatible candidates,
which one is most similar?

These two computations complement one another.

Neither replaces the other.

8. Relation to Differential Trees

Differential Trees naturally organize structural differences.

Most branches correspond to ordinary variations.

Only a few branches determine major semantic transitions.

These branches represent decisive structural deltas.

Recognition can therefore use Differential Trees to:

- locate the decisive branch,
- identify the controlling attribute,
- classify the decision side,
- construct structurally valid candidate sets.

9. Engineering Implications

Future AI systems should explicitly detect decisive structural deltas before making important decisions.

Applications include:

- code review,
- software verification,

- legal analysis,
- medical diagnosis,
- runtime scheduling,
- safety engineering,
- certified component retrieval,
- AI agent planning.

In all of these domains, finding the decisive delta is often more important than computing another decimal place of similarity.

10. Toward Structural Recognition

The recognition process therefore becomes:

Input



Shared Structural Base



Decisive Structural Delta



Boundary Recognition



Structural Identity



Similarity Ranking



Decision

This ordering reflects the central philosophy of SRMS:

Recognition establishes identity.

Similarity ranks alternatives within that identity.

Key Takeaways

- Most structural differences are insignificant.
- A small number of structural deltas determine the entire decision.

- Decisive deltas are sparse, localized, and high-impact.
- Decision boundaries are structural rather than metric.
- Recognition must identify decisive deltas before similarity scoring.
- Differential Trees provide a natural representation for decisive structural deltas.

SRMS Principle 004

Reliable AI depends not on detecting every structural difference, but on recognizing the small set of decisive structural deltas that determine semantic identity, runtime behavior, and decision boundaries.